

CS & IT ENGINEERING



Computer Network - 2

MAC Sub-Layer

Lecture No. - 03

By - Abhishek Sir





Recap of Previous Lecture



Topic

Pure ALOHA





Topics to be Covered



Topic

Slotted ALOHA

Topic

CSMA

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
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[NAT]

[GATE-2021][2 Mark]



#Q. Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 1 Mbps ($=10^6$ bits per second). The aggregate number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 1,000 frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer) is _____.

1 sec $\longrightarrow$ 1000 frames
($T_x = 10^{-3}$ sec) $\longrightarrow$ 1 frame

Ans = 135 to 138
IITB Ans = 131 to 140

$G = 1$ $\leftarrow$ Mean no. of transmissions per frame time

$$\text{Frame Size} = 1000 \text{ bits}$$

$$\text{Bandwidth} = 1 \text{ Mbps} = 10^6 \text{ bits/sec}$$

$$\text{Frame Transmission Time } (t_x) = \frac{\text{Frame Size}}{\text{Bandwidth}} = \frac{10^3 \text{ bits}}{10^6 \text{ bits/sec}} = 10^{-3} \text{ sec}$$

$(t_x = 10^{-3} \text{ sec}) \longrightarrow 0.13533 \text{ frames transmitted successfully}$

1 sec $\longrightarrow 135.33 \text{ frames transmitted successfully}$

For Pure ALOHA,

$$S = G * e^{-2G}$$

$$S = \frac{1}{e^2}$$

0.13533

0.13717

[Put $G=1$]

$S \rightarrow$ Avg. no. of successful transmissions per frame time

Average number of frames successfully transmitted per second

= S * Aggregate number of transmissions per second

= $0.13533 * 1000 \text{ frames/sec}$

= $135.33 \text{ frames/sec}$



Topic : Pure ALOHA



$$P = \frac{S}{G}$$

$$S = G * e^{-2G}$$

Maximum throughput (**S**) can achieve at $G = 0.5 = \frac{1}{2}$

$$\text{Maximum throughput (S)} = 1/2e = 0.1839 = 18.39 \%$$



Topic : Slotted ALOHA



-> To improve the efficiency in Pure ALOHA

-> Divide the time into equal size time-slots
[Slot time = One frame transmission time]

-> Nodes are synchronized

-> Whenever a node has packet to transmit,
it will start transmission only at begin of slot



Topic : Slotted ALOHA



Vulnerable time

-> Time duration in which no transmission should done to avoid collision

$$\text{Vulnerable time} = \text{Frame Transmission Time} = T_x$$



Topic : Slotted ALOHA



[Vulnerable time
 $= t_x$]

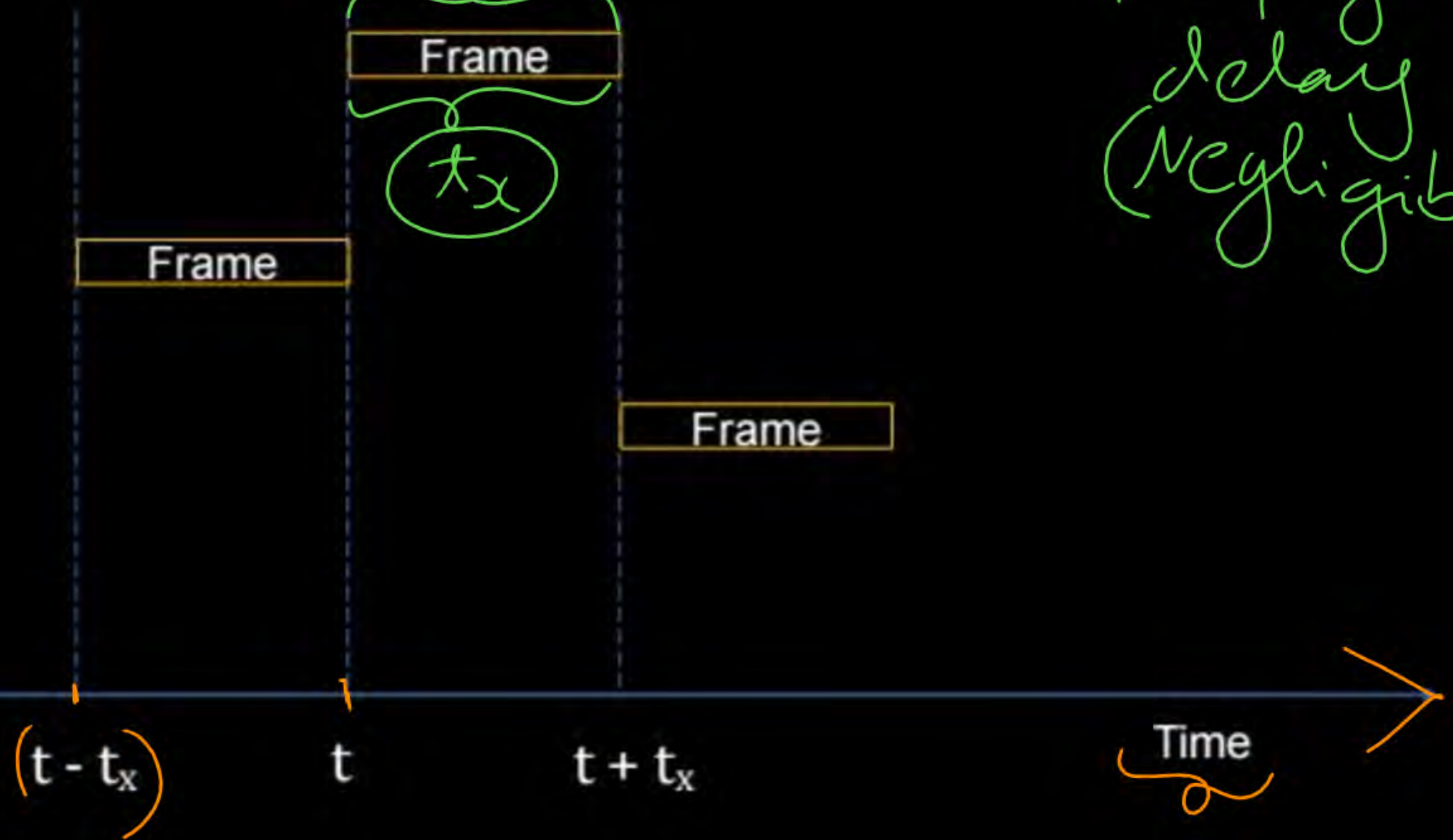
* Ignore
propagation
delay
(Negligible)

A

B

C

D



Example 4 :-

[NAT]



#Q. Consider a network using the slotted ALOHA medium access control protocol, where each frame is of length 4000 bytes. The channel transmission rate is 1 Mbps. Calculate vulnerable time (in milliseconds) ?

Solution 4 :-

$$\text{Frame Size} = 4000 \text{ Bytes} = 32 \times 10^3 \text{ bits}$$

$$\text{Bandwidth} = 1 \text{ Mbps} = 10^6 \text{ bits/sec}$$

$$\text{Frame Transmission Time } (t_x) = \frac{\text{Frame Size}}{\text{Bandwidth}} = \frac{32 \times 10^3 \text{ bits}}{10^6 \text{ bits/sec}} = 32 \text{ ms}$$

For Slotted ALOHA :

$$\text{Vulnerable time} = \text{Frame Transmission Time } (t_x) = 32 \text{ ms}$$

$$\boxed{\text{Ans} = 32}$$



Topic : Slotted ALOHA



According to Poisson's distribution

$$P = e^{-G}$$

$$S = G * P$$

$$S = G * e^{-G}$$

$\rightarrow G$, vulnerable time

V.T. $\Rightarrow 2t_x$, Pure ALOHA
 $P = e^{-2G}$

V.T. $\Rightarrow t_x$, Slotted ALOHA
 $P = e^{-G}$

Example 5 :-

[NAT]



#Q. Consider a network using the slotted ALOHA medium access control protocol, where each frame is of length 4,000 bits. The channel transmission rate is 8 Mbps. The aggregate number of transmissions (including retransmissions) across all the nodes with a rate of 2,000 frames per second. Throughput is defined as the average number of frames successfully transmitted per frame time. The throughput of the network is _____.

$$\begin{aligned} t_x &= \frac{4000 \text{ bits}}{8 \text{ Mbps}} \\ &= \frac{4 \times 10^3 \text{ bits}}{8 \times 10^6 \text{ bits/sec}} \\ &= \frac{1}{2} \times 10^{-3} \text{ sec} \end{aligned}$$

$$1 \text{ sec} \longrightarrow 2000 \text{ frames}$$

$$t_x = \frac{10^{-3}}{2} \text{ sec} \longrightarrow \frac{10^{-3}}{2} \times 2 \times 10^3 \text{ frames}$$

$$t_x \longrightarrow 1 \text{ frames}$$

$$\boxed{G = 1}$$

Slotted ALOHA:-



$$S = G * e^{-G}$$

$$[put\ G=1]$$

$$S = \frac{1}{e}$$

$$0.3678$$

$$0.37037$$

$$[Ans = 0.367 \text{ to } 0.37]$$



Topic : Slotted ALOHA



$$S = G * e^{-G}$$

For what value of G throughput S is maximum ?

$$\frac{dS}{dG} = 0$$



Topic : Slotted ALOHA



$$\frac{d}{dG} [G * e^{-G}] = 0$$

$$G * \frac{d}{dG} e^{-G} + e^{-G} * \frac{d}{dG} G = 0$$

$$G * e^{-G} * (-1) + e^{-G} = 0$$

$$e^{-G} - G * e^{-G} = 0$$

$$e^{-G} * [1 - G] = 0$$

$$1 - G = 0$$

$$\boxed{G = 1}$$



Topic : Slotted ALOHA



$$S = G * e^{-G}$$

Maximum throughput (S) can achieve at $G = 1$

[put $G = 1$]

$$S = \frac{1}{e}$$



Topic : Slotted ALOHA



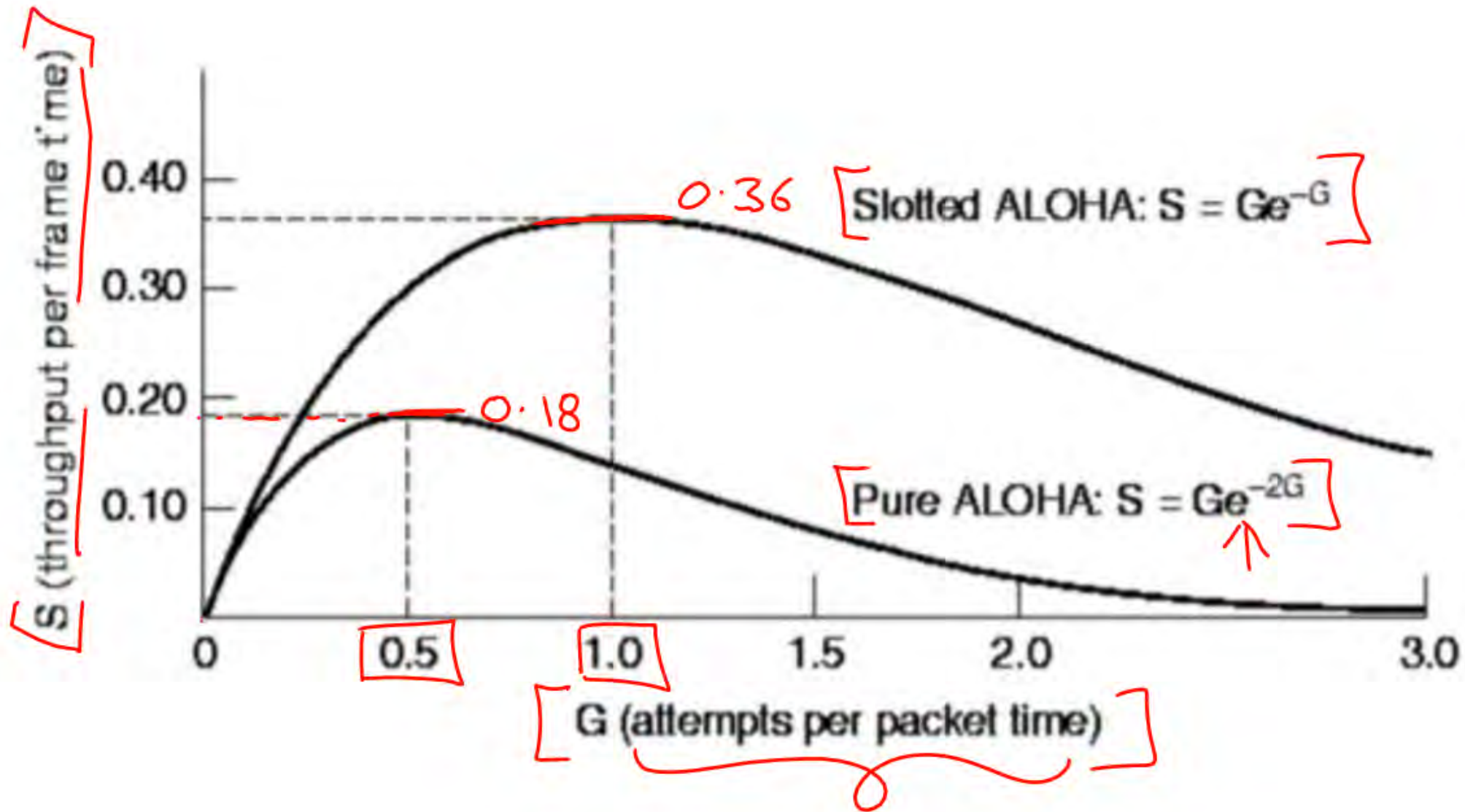
$$S = G * e^{-G}$$

Maximum throughput (S) can achieve at $G = 1$

$$\text{Maximum throughput} = 1/e = 0.3678 = 36.78\%$$



Topic : Efficiency



Example 6 :-

[MCQ]



#Q. Consider n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that particular one station transmits in a given time slot?

- A $(1 - p)^{(n-1)}$
- ☒ B $p * (1 - p)^{(n-1)}$
- C $n * p * (1 - p)^{(n-1)}$
- D $1 - (1 - p)^{(n-1)}$

Ans: B

H_A

H_B

H_C

H_D

$$P \times (1 - P) \times (1 - P) \times (1 - P)$$

$$P \times (1 - P)^{(n-1)} \leftarrow \text{Throughput of Host A}$$

#Q. There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that ONLY one station transmits in a given time slot?

- A $(1 - p)^{(n-1)}$
- B $p * (1 - p)^{(n-1)}$
- ☒ C $n * p * (1 - p)^{(n-1)}$
- D $1 - (1 - p)^{(n-1)}$

Throughput of Channel
= Throughput_A + Throughput_B
+ + Throughput_D

$$= n * P * (1 - P)^{(n-1)}$$

Ans: C



Topic : Slotted ALOHA Efficiency



- > Suppose N nodes with many frames to transmit
- > Each node transmits in slot with probability p

Probability that particular node has success in a given slot = $p * (1 - p)^{(N-1)}$
(Throughput of Host)

Probability that any node has success in a given slot = $N * p * (1 - p)^{(N-1)}$
(Throughput of Channel)



Topic : Slotted ALOHA Efficiency



$$\text{Throughout of Channel (A)} = N * p * (1 - p)^{(N-1)}$$

For what value of p throughput A is maximum ?

$$\frac{dA}{dp} = 0$$

$$\frac{d}{dp} [N \times p \times (1-p)^{(N-1)}] = 0$$

$$N \times \left[p \times \frac{d}{dp} (1-p)^{(N-1)} + (1-p)^{(N-1)} \times \frac{d}{dp} p \right] = 0$$

$$\left[p \times (N-1) \times (1-p)^{(N-2)} \times (-1) + (1-p)^{(N-1)} \right] = 0$$

$$(1-p)^{(N-1)} \times \left[p \times (N-1) \times \frac{1}{(1-p)} \times (-1) + 1 \right] = 0$$

$$1 - \frac{p \times (N-1)}{(1-p)} = 0$$

$$1 - \frac{p \times (N-1)}{(1-p)} = 0$$

$$p \times (N-1) = 1-p$$

$$pN - p = 1 - p$$

$$pN = 1$$

$$\boxed{p = \frac{1}{N}}$$



Topic : Slotted ALOHA Efficiency



$$\text{Efficiency} = N * p * (1 - p)^{(N-1)}$$

Maximum efficiency can achieve at $p = 1/N$

$$\text{Maximum efficiency} = \underline{1/e}$$

Throughput of Channel

$$\begin{aligned} &= N * p * (1 - p)^{(N-1)} \\ &= N * \frac{1}{N} * \left(1 - \frac{1}{N}\right)^{(N-1)} \\ &= \lim_{N \rightarrow \infty} \left(1 - \frac{1}{N}\right)^{(N-1)} \\ &= \underline{\frac{1}{e}} \end{aligned}$$

[MCQ]

IIT → H.W.

[GATE-IT-2004] [2 Mark]



#Q. Consider a simplified time slotted MAC protocol, where each host always has data to send and transmits with probability $p = 0.2$ in every slot. There is no backoff and one frame can be transmitted in one slot. If more than one host transmits in the same slot, then the transmissions are unsuccessful due to collision. What is the maximum number of hosts which this protocol can support, if each host has to be provided a minimum through put of 0.16 frames per time slot?

- A 1
- B 2
- C 3
- D 4

[NAT]

11T-K, H.W.

[GATE-2015][2 Mark]



- #Q. Consider a LAN with four nodes S1, S2, S3 and S4. Time is divided into fixed-size slots, and a node can begin its transmission only at the beginning of a slot. A collision is said to have occurred if more than one node transmit in the same slot. The probabilities of generation of a frame in a time slot by S1, S2, S3 and S4 are 0.1, 0.2, 0.3 and 0.4, respectively. The probability of sending a frame in the first slot without any collision by any of these four stations is _____.



Topic : CSMA

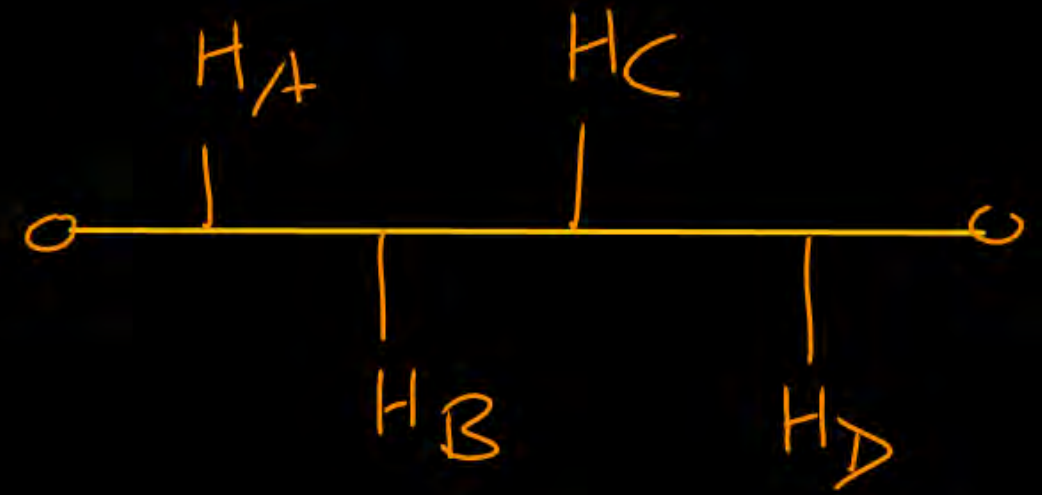


→ Carrier Sense Multiple Access

→ Sense before transmit
[Sense the channel, before transmission]

=> if channel sensed idle : "transmit entire frame"

=> if channel sensed busy : "defer transmission"





Topic : CSMA



Different variations of CSMA protocols :

- i. 1 – Persistent CSMA
- ii. Non – Persistent CSMA
- iii. p – Persistent CSMA



Topic : 1-persistent CSMA



→ Sense before transmit

→ if channel sensed idle
transmit entire frame immediately (with probability 1)

→ if channel sensed busy
sense the channel continuously (till it become idle)

[Ethernet $\Rightarrow$ 1-persistent CSMA]

Better
In case of
low traffic
To improve
performance



Topic : Non-persistent CSMA



→ Sense before transmit

→ if channel sensed idle
transmit entire frame immediately

→ if channel sensed busy
Wait a random amount of time and then sense the channel

It do not sense the channel continuously when channel is busy.
[Unlike 1 - Persistent CSMA]

(Better
In heavy
traffic)
(Reduces
no. of collision)

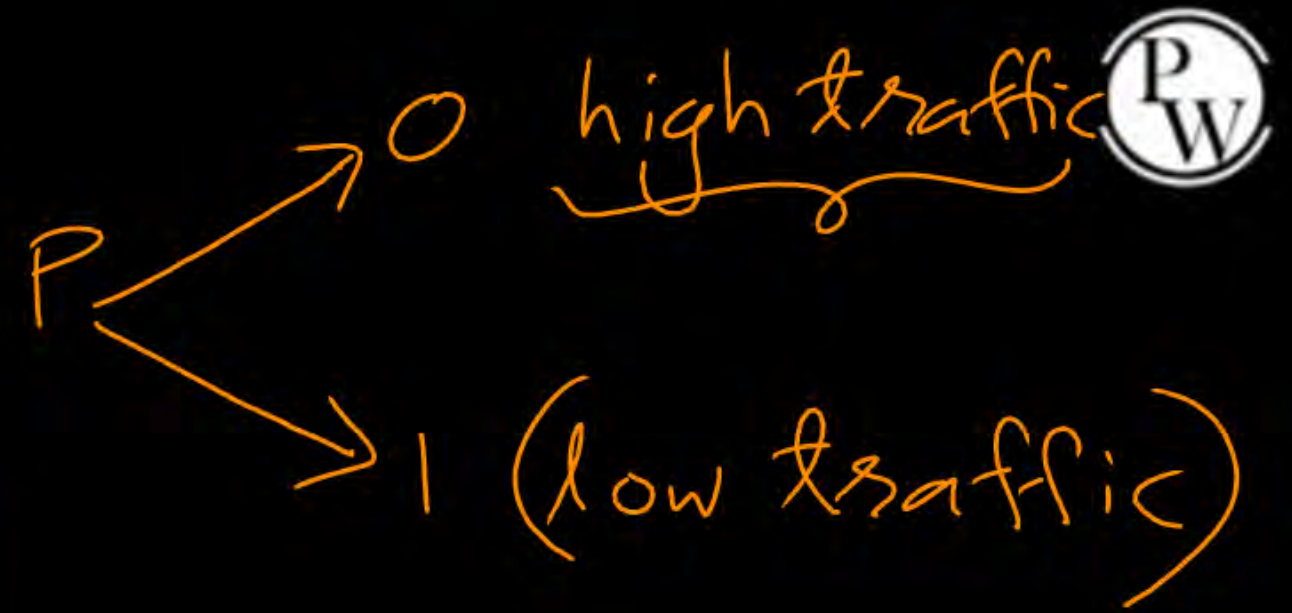


Topic : p-persistent CSMA

→ Sense before transmit

→ if channel sensed idle
transmit the frame with probability p

→ if channel sensed busy
sense the channel continuously (till it become idle)





2 mins Summary



Topic

Slotted ALOHA

Topic

CSMA



THANK - YOU

